

SEMATECH and Collaborative Research: Lessons in the Design of High-Technology Consortia

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Abstract

This article reviews the experience of SEMATECH as a model for high-technology research consortia. SEMATECH's original aims of developing next-generation manufacturing technology proved hard to achieve, and the program has refocused on generic technology and the equipment industry infrastructure. Though more modest, these new objectives have produced significant tangible results. The study considers the reasons for the change and implications for consortium design. This is contrasted with the history of other major collaborative research programs in Japan, Europe, and the United States.

INTRODUCTION

Since its establishment in 1987, SEMATECH (the Semiconductor Manufacturing Technology Consortium) has received considerable attention from U.S. policymakers and managers. The consortium has been endorsed by the Clinton administration as a model of government-industry cooperation in supporting research in "critical technologies."¹ Other industry-led consortia have been proposed or established in areas ranging from electric automobiles (the Advanced Battery Consortium) to high-definition displays and other semiconductor industry products (the proposed "multi-chip module foundry" consortium) to aircraft development and production (the proposed Aerotech consortium). Contrary to the predictions of Cohen and Noll [1992],

¹ "SEMATECH, an industry consortium created to develop semiconductor manufacturing technology . . . can serve as a model for federal consortia funded to advance other critical technologies. Programs will be encouraged in the development of a new automobile, new construction technologies, intelligent control and sensor technologies, rapid prototyping, and environmentally-conscious manufacturing" [Clinton and Gore, 1993].

SEMATECH has been followed by a number of other U.S. technology development consortia funded from public and private sources.²

With these concerns in mind, we attempt to derive some lessons from SEMATECH for the design of such consortia in other industries. Although unusual in the U.S. context, SEMATECH is not unique—both the Japanese and Western European governments have sponsored civilian technology consortia since the 1960s. With and without federal funding, consortia have sprung up in other U.S. industries since 1980. Our discussion of SEMATECH compares it with these other domestic and international consortia to develop some insights into the relevance of collaborative R&D programs in other U.S. industries.

Changes in SEMATECH's role and research agenda since its foundation have important implications for this discussion. Since its foundation, SEMATECH has shifted from "horizontal" research cooperation to "vertical" collaboration between its members—major users of semiconductor process equipment and materials—and the U.S. suppliers of these goods, a structure that may be applicable to other U.S. industries. In many respects, SEMATECH now resembles an industry association, diffusing information and best-practice techniques, setting standards, and coordinating generic research. Like many Japanese cooperative research projects, SEMATECH now is concerned as much with technology diffusion as with the advancement of the technological frontier [Hane, 1994, makes a similar point]. The SEMATECH research agenda emphasizes near-term results, a characteristic that it shares with other industry-led consortia in the United States and elsewhere. Nevertheless, SEMATECH's vertically oriented research agenda has not met the goals of all its original members, several of whom have left the consortium. Moreover, SEMATECH's efforts have been insufficient to prevent several equipment firms from exiting the industry.

One of the most serious challenges in any analysis of this and other federal technology policies is evaluation. The politically relevant timetable for such evaluation is often far shorter than the period of time needed for the realization of the economic effects of consortia research. The specific aims of research consortia often change during their operation, as was the case with SEMATECH. Quantitative data on the operations and economic effects of this consortium also are difficult to obtain, because of the competitively sensitive nature of much of the information and because of the short time period during which the consortium has been in operation. Our analysis of SEMATECH draws on the extensive coverage of the consortium in the trade press, studies sponsored by various federal agencies, research in the SEMATECH archives and project records, and interviews with SEMATECH staff, assignees from member firms, and managers of U.S. semiconductor equipment firms.³ In addition, we draw on previous work on collaborative

² The analysis by Cohen and Noll [1992] builds on their previous study of federal civilian technology programs [1991]. But Cohen and Noll appear to overstate the importance of the "results-focused" research agenda of publicly funded civilian technology programs in arguing that they are fundamentally incompatible with the distributive politics of the U.S. Congress. Moreover, Cohen and Noll may understate the distributive politics of large-scale publicly funded basic research programs in the United States [Mowery, 1992a].

³ We conducted more than 25 interviews with these individuals during July 1992–June 1993, and compiled detailed histories of eleven SEMATECH projects. We were assisted in these efforts

R&D in Japan and Western Europe by the authors of this article and other scholars to develop our comparative analysis. We do not aspire to a definitive verdict on SEMATECH in this study, but our discussion of evaluation criteria yields some implications for the future aims and agenda of the consortium, to which we return briefly in our conclusions.

DESIGN AND POLICY ISSUES FOR R&D CONSORTIA

The economics of collaborative research have received considerable attention in recent studies [Katz and Ordover, 1990; Mowery and Rosenberg, 1989; Bozeman et al., 1986]. The hypothesized advantages of collaboration in research include the ability of participating firms to lower costs and spread risks, reduced duplication in their R&D investments, and the exploitation of economies of scale in the R&D process. Research consortia also may be able to appropriate more of the returns to their R&D investments and internalize more of the interfirm "spillovers" that discourage R&D investment by individual firms (because of the hypothetically greater ability in appropriating its returns, this factor favors collaboration in basic research).

These hypothesized advantages of collaborative research have not been subjected to detailed empirical scrutiny, however, and some may be exaggerated or nonexistent. The output of cooperative research must be absorbed by the participant firms and transformed into commercially relevant knowledge. In order to exploit externally performed research, whether this research is performed in a multifirm consortium, a federal laboratory, or a university, participant firms must invest in the creation of in-house expertise [Mowery, 1983; Cohen and Levinthal, 1989]. Some duplication of the in-house research investments of firms thus is likely even among participants in research consortia.⁴

Reduced duplication of research projects also creates a potential disadvantage. Diversity in research projects provides an important hedge against the possibility that any single research project will be fruitless [Nelson, 1961]. Yet if all or most firms in an industry participate in a cooperative research scheme, reducing the number of independent lines of inquiry, then such diversity and its attendant advantages may be lost. Although it is not a necessary result of research cooperation, such centralization could lower the productivity of an industry's or a nation's research investment.

How common are the R&D economies of scale in R&D that are widely cited as a source of efficiency gains through research collaboration? The

by Mike Farrell, Ansel Goldgar, and John Pankratz of the SEMATECH staff, none of whom shares responsibility for the conclusions of this analysis.

⁴ Moreover, to the extent that basic research requires a larger investment in this in-house "absorption mechanism" [see the report of the Committee on Industry and Trade, 1929], participants in cooperative research may react to the high costs of exploiting collaborative basic research by shifting the collaborative research agenda away from basic research [Mowery and Rosenberg, 1989]. The experience of the industry-funded Microelectronics and Computer Technology Corporation (MCC) illustrates this tendency. Since its foundation, MCC has shifted its research agenda somewhat from basic to near-term research, and now relies more heavily on member firms for the performance of this research. The shift in MCC's research to smaller-scale, shorter-term projects reflects the desire of members to show measurable returns on their contributions, as well as the difficulties of translating basic research into products [Lineback, 1987a, 1987b].

average costs of research experiments at large, specialized research facilities, such as wind tunnels or advanced instrumentation, will decline with more intensive use. But there exists no evidence to indicate that all industrial R&D is characterized by such economies of scale [Fisher and Temin, 1973], nor are there indicators that enable one to identify the industries or technologies characterized by R&D scale economies.

Finally, economic analyses of cooperative research often overlook the extent to which appropriability concerns manifest themselves within a consortium. Even a "precommercial" R&D consortium typically involves the sharing of sensitive information and know-how among competing firms. The same fear of knowledge spillovers that often causes firms to underinvest in R&D in the absence of a consortium may also lead them to resist knowledge-sharing within a consortium [Evan and Olk, 1990]. These concerns over appropriability and spillovers, coupled with tensions arising from overall differences and conflicts among the objectives of consortia members, may impede their performance, increase costs, and even result in dissolution.⁵

The complexity of consortium design and management is attributable to at least three challenges: definition of the research agenda and choice of projects; transfer of research results to participants; and adaptation to change in the economic and technological environment. Defining a consortium's research agenda is difficult because of appropriability concerns and divergent member firm objectives. Firms may be unwilling to entrust the development of their core technologies to the consortium or to risk the leakage of key proprietary technological and strategic information. These problems, particularly divergence among members' objectives, become more severe as the number of firms in a consortium increases.

Consortia in the United States, Western Europe, and Japan have developed different solutions to this challenge. Those involving a large number of participants (more than 30), including such European consortia as ESPRIT and Alvey, often adopt a decentralized structure and research agenda, funding a number of independent projects that each involve a small number of firms, perhaps as few as two or three.⁶ This structure facilitates control of the research by individual firms, and reduces knowledge spillovers among participants. The U.S. MCC consortium began with a complex research agenda that emphasized selective collaboration within a broad umbrella organization, but has evolved toward a decentralized structure that resembles that of ESPRIT in some respects.⁷ Another U.S. consortium, NCMS, has developed a similar structure.

Recent Japanese consortia, such as the VLSI program, also display tensions between sharing and appropriating the results of research in their blend of firm-specific, individual projects and joint research facilities [Fransman, 1990; Hane, 1994]. But the large "national project" consortia, which tend to have a small number of participants (fewer than 10), exhibit greater centralization in their approach to establishing their research strategy than do the

⁵ "Consortiums are the wave of the future, but they sure are hard to manage," according to former SEMATECH COO Paul Castrucci [quoted in Smith, 1989].

⁶ A summary listing of the characteristics of some recent R&D consortia in the United States, Western Europe, and Japan may be found in Table 1. For a listing of acronyms used in our discussion, see Appendix A.

⁷ This change in its structure occurred concurrently with expansion in MCC's membership.

Table 1. Major research consortia in Japan, the United States, and Europe.

Name	Country	Dates	Budget ^a \$m	Govt share (%)	Structure	Role of managers	Central R&D site	Members	Projects	Description of research focus
VLSI	Japan	1976-80	350	40	centralized	proj. mgmt.	yes	6	—	Semiconductor R&D.manuf.
OMCS	Japan	1979-85	90	100	centralized	proj. mgmt.	yes	—	—	Optoelectronics R&D
VHSIC	USA	1980-89	900	100	centralized	admin.	no	9 ^f	—	Military semiconductor R&D
Supercomp	Japan	1981-89	130	100	centralized	proj. mgmt.	no	—	—	High speed computer R&D
FED	Japan	1981-90	40	100	centralized	proj. mgmt.	no	—	—	Semiconductor R&D
5G	Japan	1982-91	426	100	centralized	proj. mgmt.	yes	10	—	Computer software, AI
Alvey	UK	1983-88	500	50	umbrella	admin.	no	127	309	Computer software, VLSI
ESPRIT I	Europe	1984-89	1800	50	umbrella	admin.	no	500	226	IT, software
ESPRIT II	Europe	1988-93	3800	50	umbrella	admin.	no	—	—	IT, software
Eureka	Europe	1985-96	7700	50	umbrella	admin.	no	—	297	Electronic technology
RACE	Europe	1985-96	3000	50	umbrella	admin.	no	—	—	Telecommunications
JESSI	Europe	1989-96	4000	50	umbrella	admin.	no	—	70	Semiconductor manuf. R&D
MCC	USA	1983-1986	80 pa ^b	0 ^c	mixed ^e	proj. mgmt.	yes	50	—	Semiconductor basic R&D
NOMS	USA	1986-	150 pa	50 ^d	umbrella	proj. mgmt.	no	180	—	Manufacturing processes
SEMATECH I	USA	1987-92	1000	50	centralized	proj. mgmt.	yes	11	—	Semiconductor manuf. R&D
SEMATECH II	USA	1993-	200 pa	50	centralized	proj. mgmt.	yes	11	—	Semiconductor manuf. R&D

Note: VLSI Budget: ¥73.7bn (MITI contribution: ¥29.1bn); OMCS: ¥16bn; Supercomputer: ¥23bn; FED: ¥7.6bn 1981-86 only [Fransman, 1990, p. 182]; 5G ¥24bn [\$133m] 1982-87 only [Fransman, 1990, p. 228]; Alvey: £350m; ESPRIT I: ECU 1.5bn; ESPRIT II: ECU 3.2bn; Eureka: ECU 6.5bn [Watkins, 1991]; JESSI: ECU 3.5bn original plan, since reduced to about ECU 2.5bn [Watkins, 1991]; MCC: \$300m 1983-90; RACE: ECU 550m first 5 years only. All ECU translated at \$1.20 to ECU 1; Yen at ¥210 to \$1 for VLSI and ¥180 to \$1 for more recent Japanese consortia.

^a Total amounts (government plus industry).

^b pa: per annum. Reflects the most recent annual budget figure.

^c While MCC gets no lump sum government funding, the consortium competes for grants.

^d Combination lump-sum and competitive grant funding.

^e Began centralized, has evolved toward umbrella form.

^f VHSIC participants total 14 firms with joint contractors.

Western European consortia. In previous Japanese consortia, this organizational design was palatable to member firms because all of them were a considerable distance behind the world technological frontier.⁸ As the technological capabilities of these firms have improved, forging cooperative research agreements among them has become more difficult [see Odagiri and Goto, 1993].⁹

A related issue in the definition of the research agenda for R&D consortia is the time horizon of the research. Despite their frequent promotion in the United States as a solution to "myopic" R&D management, consortia that do not include universities or national laboratories rarely focus on long-term, fundamental research. Instead, successful consortia in the United States, Western Europe, and Japan typically define a relatively short-term research agenda. Indeed, the temporary nature of most Japanese consortia makes it difficult for them to undertake long-term research.

The second challenge facing consortia design involves the transfer of technology to participants. This process operates more successfully when cooperative research activities are complemented by significant in-house investments in related research by program participants. The size of this parallel investment and managerial commitment to the projects will depend on the selection of projects that are central to the firms' broader commercial and research interests. Technology transfer also is aided by rotation of member firms' personnel through collaborative or external research establishments.

A third important challenge for consortia is adjusting to change in their environment or research agenda. The difficulties of negotiating a consortium research agenda in the first place mean that gaining agreement among members on its revision may be impossible. Even if agreement can be reached, the consortium's original structure or facilities may prove unsuited to the new agenda. The organizational structure of a consortium also affects the feasibility of revisions in its research agenda—a strong administrative structure, rather than a weak "secretariat," appears to be necessary. Some consortia, such as the Alvey and ESPRIT programs in the United Kingdom and Western Europe, have avoided flexibility in the research agenda altogether. For SEMATECH, such flexibility has been an important strength, albeit one that has come at some cost.

THE FOUNDATION AND STRUCTURE OF SEMATECH

Structure and Objectives

The origins of SEMATECH can be traced to late 1986 and early 1987, when Japanese firms came to dominate the global market for semiconductor mem-

⁸ Indeed, many Japanese cooperative research programs may be viewed as important components of a technology adoption policy, and some scholars have argued that their importance will decline in the Japanese economy in the future [Sigurdson, 1986; Okimoto, 1987; Fransman, 1990; Mowery and Rosenberg, 1989; Odagiri and Goto, 1993].

⁹ See Samuels [1987] and Fransman [1990], who notes that "the perceived threat from IBM as the dominant player in the international computer industry, helped to galvanize a significant degree of solidarity and cooperation in Japan. It was obvious that Japanese computer producers were under far greater competitive pressure from IBM than they were from each other. The dominance of IBM therefore helped create the conditions for a closer degree of cooperation between the Japanese companies" (p. 59).

ory chips. Two reports influenced its creation. The Defense Science Board (DSB), a committee of scientists and engineers that advises the Undersecretary of Defense for Research and Engineering, issued a report in 1987 asserting that U.S. semiconductor firms were falling behind Japanese firms in high-volume manufacturing process technologies for chips. The DSB argued that this growing gap in process technology had important national security implications. According to the report, a portion of the U.S. "defense industrial base" that had been spawned during the 1950s and 1960s by defense demand for its products now depended for its technological and economic viability on its fortunes in the commercial market. U.S. defense agencies accordingly had a vital stake in strengthening the commercial technology of this industry. The DSB recommended the creation of a manufacturing facility, jointly owned by government and industry, to produce semiconductor components [McLoughlin, 1992].

The DSB report was released shortly before a report by the Semiconductor Industry Association (SIA) that recommended the establishment of an industry-wide consortium that would conduct research, rather than manufacturing chips. In May 1987, a 14-member committee of the SIA formulated a proposal for a research consortium that called for matching federal contributions.¹⁰ These 14 companies, which accounted for over 80 percent of U.S. semiconductor component manufacturing capacity, were the founding members of SEMATECH.

SEMATECH is funded from public and private sources. The Defense Advanced Research Projects Agency (DARPA, renamed ARPA in 1992) of the U.S. Defense Department has provided \$100 million annually, matching the member firms' combined annual contribution, and the state of Texas provides modest additional funding.¹¹ The consortium's staff includes many employees of member firms, in order to improve the transfer of technology from SEMATECH to member firms. In 1992, 225 of the 722 personnel employed at SEMATECH were assignees from member firms—most of these individuals were members of the consortium's 300-member professional research staff. SEMATECH conducts research in its own facilities as well as at member firms and at equipment suppliers. SEMATECH also contributes funds to the Semiconductor Research Corporation, an industry-financed organization that finances research at U.S. universities [SEMATECH, 1991]. SEMATECH is not open to firms that are not headquartered in the United States or that are substantially controlled by non-U.S. citizens.

Simultaneously with the foundation of SEMATECH, U.S. semiconductor materials and equipment (SME) suppliers formed SEMI/SEMATECH, established in September 1987 to facilitate linkages between U.S. SME suppliers and SEMATECH. SEMI/SEMATECH currently has 135 members (out of

¹⁰ The working group included one senior manager from each of the following firms: Advanced Micro Devices; AT&T; Digital Equipment Corporation; Harris Corporation; Hewlett-Packard Company; Intel Corporation; IBM; LSI Logic; Micron Technology; Motorola; National Semiconductor; NCR; Rockwell International; and Texas Instruments.

¹¹ Membership dues are 1 percent of semiconductor sales (including the value of production for internal use by semiconductor producers such as IBM), with a minimum of \$1 million and a maximum of \$15 million, and smaller semiconductor firms have found it expensive to participate in the consortium. In 1991, SEMATECH's operating expenditures amounted to \$233 million; the excess over \$200 million represented money unspent from the previous year.

roughly 800 U.S. firms in the SME industry) who account for more than 85 percent of U.S. SME sales. The organization, which has 11 full-time employees headquartered in a wing of SEMATECH's Austin facility, is funded by membership dues and by fees generated through sponsorship of conferences.¹² Dues are based on company revenue and start at less than \$5000. The president of SEMI/SEMATECH sits on the board of SEMATECH, and the boards of SEMI/SEMATECH and SEMATECH meet jointly once each quarter.¹³

Evolution of SEMATECH Objectives

Originally intended to provide a research facility for member firms to collaborate on projects to improve their semiconductor manufacturing process technology, SEMATECH now focuses on strengthening the semiconductor manufacturing equipment (SME) industry. SEMATECH's original objectives underpinned the decision to build a large-scale fabrication facility in Austin. Fab One, completed in 1989 in the record time of 32 weeks, was a triumph of collaborative effort [U.S. GAO, 1989]. But once the ambitious construction project had been completed, SEMATECH had difficulty developing a research agenda that could exploit it. Members questioned whether the development of advanced manufacturing processes was an appropriate objective for an industry consortium.¹⁴ Process technology expertise is central to the competitive advantage of individual semiconductor manufacturers, and member firms were reluctant to share such sensitive information. The sophistication of the manufacturing technology of SEMATECH member firms also differed considerably, raising the danger that some firms could "free ride" on the contributions of technology leaders.

In response to these controversies, SEMATECH altered its research agenda to one that sought to improve the technological capabilities of U.S. suppliers of semiconductor manufacturing equipment [Katz and Ordovery, 1990; U.S. CBO, 1990], and to strengthen "vertical" cooperation between U.S. suppliers and U.S. users of semiconductor process equipment.¹⁵ This research focus potentially benefits all members without threatening their proprietary capabilities, and thereby may reduce the appropriability problems associated with the original research agenda. The new research agenda has shifted the consortium's focus from the development of a complete state-of-the-art production process in its Austin facility to knowledge diffusion and technology transfer. In the words of William Spencer, current SEMATECH CEO, "We can't develop specific products or processes. That's the job of the member companies. SEMATECH can enable members to cooperate or compete as they see fit" [Burrows, 1992, p. 58].

This revised research strategy is illustrated by SEMATECH's announcement in January 1993 that it had achieved its goal of manufacturing 0.35

¹² SEMI/SEMATECH's 1991 operating budget was approximately \$1.6 million.

¹³ SEMI/SEMATECH should not be confused with the trade association SEMI (Semiconductor Equipment and Materials International). SEMI has over 1500 member firms, of which only 800 are U.S.-based. While the members of SEMI/SEMATECH are generally also members of SEMI, the two organizations are separate and have distinct roles and objectives.

¹⁴ A similar question is raised in Mowery and Rosenberg [1989, p. 271].

¹⁵ According to Levy and Samuels [1991], 80 percent of collaborative research agreements in Japanese industry, few of which are publicly financed, involve "vertical" relationships (p. 123).

micron line-width integrated circuits. Rather than developing a "recipe" and associated processes for the production of specific components that could be readily adopted by its member firms, the consortium had succeeded in developing, integrating, and operating equipment that could be used to produce a variety of 0.35 micron integrated circuits. In other words, the 0.35 micron line-width process technology was generic, rather than product-specific or firm-specific. According to a spokesman for AMD, "What SEMATECH has done is an outstanding achievement, but it is not realistic to think a member of SEMATECH could take the process in its entirety and produce product with the same equipment and the same alignment" [Cassell and Dunn, 1993].

The shift in its research agenda was associated with changes in SEMATECH's intellectual property policies. Originally, the results of SEMATECH research were licensed to member firms on an exclusive basis for two years, after which they were to be made available to U.S. firms at nominal royalty rates. Its new emphasis on the supplier industry led SEMATECH in 1991 to relax its prohibition on the disclosure of SEMATECH-funded equipment advances to nonmember firms; instead, member firms now have priority in ordering and receiving such equipment [U.S. GAO, 1992a]. Once SEMATECH member demand is satisfied, equipment suppliers can sell to any customer.¹⁶ The change in disclosure policies benefited U.S. equipment firms, and also addressed criticism from some semiconductor manufacturers that SEMATECH was exploiting general tax revenues to benefit member companies at the expense of nonmembers. This shift in policy has some costs for SEMATECH members, since equipment firms participating in SEMATECH projects now can sell products embodying the results of this work to all customers, including foreign firms. Still, SEMATECH participants retain important advantages because of their lead time and greater familiarity with new equipment developed under SEMATECH sponsorship. Assignees gain familiarity in the use of new technology and, by influencing the evolution of equipment technology, may improve its fit with their needs.

SEMATECH provides a central funding and testing organization for equipment development and "qualification" (demonstration of their performance specifications), which is less expensive than individual prototyping or testing. By reducing duplication among semiconductor manufacturers in the development and qualification of new tools, the consortium can lower the costs of adopting new process equipment.¹⁷ Decreasing the fragmentation of product designs and increasing knowledge of equipment maintenance, software, and operation should facilitate the adoption of U.S. manufacturing equipment by SEMATECH members.

The shift in the SEMATECH research agenda has not eliminated all conflicts among its members. ARPA, the federal agency funding SEMATECH, ex-

¹⁶ For some sophisticated equipment, a supplier may only be able to produce a handful of units per year. In such a case, SEMATECH members are likely to purchase most or all of the available product.

¹⁷ Leyden and Link [1992] present some preliminary estimates of resource savings associated with similar activities performed by the National Institute of Standards and Technology. They find that such standardization activities yield social returns exceeding 100 percent. Interviews with some managers of SME firms, however, suggested that a number of SEMATECH member firms still require extensive in-house testing and qualification of new equipment.

pressed some concern over the shift in the research agenda away from long-term, more fundamental research issues to equipment development [Katz and Ordovery, 1990], and the goals of SEMATECH member firms and Pentagon policymakers have not always been aligned throughout the consortium's brief history.¹⁸ Three firms have withdrawn from the consortium since its foundation (Micron Technology and LSI Logic withdrew in 1992, Harris Semiconductor in 1993). For both Micron Technology and LSI Logic, smaller semiconductor producers with weaker capabilities in process technology, withdrawal reflected their dissatisfaction with the consortium's decision not to pursue development of an advanced manufacturing process that all members could apply to the production of chips.¹⁹ At least two current members of the consortium, AT&T and Rockwell, have stated that they are evaluating their commitment to SEMATECH.²⁰

Current Operations

Its new research agenda has led SEMATECH to increase its spending on supplier contracts from 20 percent of its 1988 budget to roughly 50 percent in 1991, a level that may increase still further in the future. SEMATECH's interaction with SME suppliers falls into four broad categories: joint development projects; equipment improvement projects; provision of technology "road maps"; and efforts to broaden communication between suppliers and member firms. Joint Development Projects (JDP) fund the development of new equipment by suppliers, supported by member firms and SEMATECH personnel. An equipment firm that wins a SEMATECH JDP contract receives funding to cover approximately 25 percent of the project's cost and receives feedback from SEMATECH and its members throughout the project. Even suppliers that submit unsuccessful proposals meet with members of the SEMATECH proposal evaluation team to discuss the strengths and weaknesses of their proposals. The bulk of JDP development activity typically takes place at the supplier's facility. Each project is managed by a SEMATECH employee; SEMATECH engineers assigned to the project spend much of their time at the supplier, and SEMATECH hosts regular meetings between the supplier's employees and SEMATECH and member firm personnel.

A second area of SEMATECH activity is Equipment Improvement Projects (EIP) for current models of equipment. This program was introduced in response to criticism of the low reliability of U.S. semiconductor manufacturing equipment [Rice, 1989]. As with a JDP, a typical EIP involves the canvassing of member firms, a request for proposals, and the provision of feedback to

¹⁸ These divergent views of SEMATECH's agenda and role are likely to surface in other Pentagon-funded programs in "dual-use" technologies, and suggest that integrating the defense and civilian "industrial bases" and improving the economic competitiveness of U.S. high-technology industries through DOD R&D policy may prove to be difficult [Alic et al., 1992].

¹⁹ As the U.S. GAO [1992a] reported, both LSI and Micron "were more interested in pursuing SEMATECH's initial strategy to improve yields and reduce costs through better manufacturing methods than in working with suppliers to develop next-generation equipment" (p. 12).

²⁰ In late December 1993, AT&T submitted notification to SEMATECH that it wished to retain the option of leaving the consortium in two years [Dorsch, 1994]. AT&T's action does not constitute a formal notification of a decision to withdraw from SEMATECH, but it indicates that senior management is giving the possibility serious consideration. Rockwell submitted a similar notification to SEMATECH in late 1992.

all suppliers that submit proposals. In SEMATECH's early years, EIPs were typically located at its central research facility, but these projects now are more frequently sited at member firms or at suppliers. A SEMATECH engineering team manages each equipment improvement project from start to finish. The results of these projects are relayed through a variety of formal and informal mechanisms to member firms, who can improve their utilization or maintenance techniques for such equipment.²¹ In some cases, SEMATECH has purchased equipment from EIP participants and rented it to member firms at low rates; firms renting such equipment provide data on its operation under commercial-scale operating conditions to other SEMATECH members and to the supplier firm.

Both JDPs and EIPs include substantial efforts to "qualify" the new or improved equipment. Qualification is usually conducted at SEMATECH's Austin facility by both SEMATECH and supplier personnel, although qualification at member firm or supplier sites has been increasing. Qualification of new or improved equipment, which typically involves running the equipment under both normal production and "stress" conditions,²² takes approximately six to nine months. These qualification projects provide member firms with the information necessary for purchasing decisions and yield feedback on the product to the supplier. By disseminating test results to the member firms, SEMATECH reduces the amount of independent testing required of member firms that purchase new or improved equipment. Although SEMATECH members are under no obligation to purchase the supplier's equipment, suppliers' access to SEMATECH members during contract execution improves their chances of winning purchase orders, especially in projects that involve work at a member firm's production facilities. Interviewees at SEMATECH and elsewhere repeatedly noted that these contracts most frequently benefited both the supplier and user firm through the latter placing orders for equipment in which it had greater confidence.

The third SEMATECH activity is the provision of "road maps" of the anticipated path of technological innovation in the semiconductor industry. SEMATECH managers, member firm representatives, and project managers in each research area (such as lithography, metalization, software, and manufacturing methods) meet several times a year to review project performance and discuss future requirements. The road maps set the SEMATECH research agenda and coordinate its selection of research projects. They suggest goals and timetables for achieving new feature line-widths, provide information to suppliers concerning expected purchasing cycles, and facilitate planning by SME suppliers. These industry-wide forecasting activities may facilitate "systemic innovation" [Teece, 1986]. For example, a masking material firm normally may be unwilling to invest in development of a new resist until a lithographic equipment producer develops its new machine. By developing

²¹ A SEMATECH equipment-improvement project is credited with improving the mean operating time between breakdowns of Genus Inc.'s chemical vapor deposition tool from 75 hours to several hundred hours [Pollack, 1991]. Other SEMATECH work in deposition technology demonstrated the continued viability of physical deposition techniques for specific applications, information that enabled equipment firms to defer major investments in new lines of equipment (interviews with SEMATECH staff, May–October 1993).

²² "Stress testing" involves the nonstop use of the same piece of equipment to process high volumes of wafers.

and disseminating this road map, SEMATECH seeks to encourage SME suppliers to innovate concurrently rather than sequentially.²³ Needless to say, the public nature of the process that develops these road maps may also inform the development efforts of non-U.S. SME suppliers.

A related SEMATECH effort to facilitate systemic innovation entails the development of standards for equipment interfaces and for evaluation of equipment. For example, SEMATECH has initiated a Computer Integrated Manufacturing (CIM) program to standardize hardware and software interfaces across equipment, in an effort to reduce problems in the insertion of improved tools into existing production lines. SEMATECH has also developed a cost-of-ownership (COO) model to estimate the lifetime cost of ownership—including purchase, support, service, and maintenance costs—for individual pieces of equipment. The model enables members to reduce equipment costs²⁴ and provides feedback to suppliers early in the product development process. As of mid-1992, more than 100 suppliers had requested copies of the COO model to facilitate their own product development efforts [U.S. GAO, 1992a].

SEMATECH and SEMI/SEMATECH also have worked to improve communications between SME suppliers and semiconductor manufacturers. A series of meetings and conferences in Austin, including the Technical Advisory Boards (TABs), provide forums for interaction among executives of various companies and at various levels. One program in particular, Partnering for Total Quality (PFTQ), has focused on improving supplier management of costs, strategies, and financial practices to promote more fruitful relationships and development of world class capabilities. SEMATECH devotes \$5–10 million annually to PFTQ programs.

Distinctive Structural Features of SEMATECH

The original aims of SEMATECH resembled those of many Western European cooperative R&D programs (characteristics of some are summarized in Table 1), in seeking to provide government R&D assistance to an industry under pressure from more efficient foreign producers. Like these programs, as well as those of Japan, SEMATECH pursued a “precompetitive” research agenda, avoiding commercial development activities. But SEMATECH’s development of an initial research agenda that won the support of its members, as well as its shift in this agenda, depended on some structural features that differ from those of many European programs.

Its original design called for SEMATECH to carry out research in a central facility, in contrast to the European programs. Most of these consist of a collection of independent projects performed in-house by groups of member firms under an “umbrella” directorate or “secretariat.”²⁵ Rather than filling

²³ The U.S. semiconductor industry has expanded its efforts to develop common visions of the path of technological evolution in the MICROTECH 2000 planning exercise, which involves SEMATECH, the Semiconductor Research Corporation, and the Semiconductor Industry Association in the preparation of an industry-wide technical road map that will also be used by federal research agencies such as DARPA and the national laboratories [Jorde and Teece, 1993].

²⁴ For example, Intel asserts that use of the COO model has reduced the operating costs associated with a Lam Research etching tool by more than 50 percent [Burrows, 1992]. At least eight of SEMATECH’s members have adopted the COO model [U.S. GAO, 1992a].

²⁵ See Ouchi [1989] for the distinction between centralized “operating entity” consortia and “umbrella” or “secretariat” consortia.

a purely administrative role, as did the European directorates (for example, the Alvey Directorate did not staff project selection committees, but simply oversaw the disbursement of funds), SEMATECH managers were closely involved in their development projects. SEMATECH's approach also differs from that of many Japanese research associations, in which most research is conducted at member firms.²⁶

An important factor in the differences between SEMATECH and many European consortia is the fact that the design of SEMATECH and much of the impetus for its creation came largely from industry, rather than the national or regional governmental organizations that have spawned many European programs.²⁷ Substantial industry involvement (the participation of senior managers and their commitment of corporate funds) in SEMATECH's formation, management, and staffing increased the likelihood that the consortium would focus on problems viewed by industry as central to its near-term survival. Indeed, many of our interviewees argued that a number of SEMATECH's least successful projects has been undertaken at the initiative of SEMATECH staff, rather than being chosen by managers from the member firms. The consortium has intensified its efforts to involve a broader cross-section of member-firm managers in project selection and evaluation, seeking to gain support for projects from more than just the R&D organizations of member firms.

This commitment by U.S. firms to SEMATECH, along with its centralized management structure, gave the consortium sufficient autonomy to revise its research agenda in response to changing developments in the semiconductor industry. This flexibility contrasts with the fixed agenda of the European consortia, such as ESPRIT and Alvey, which committed the bulk of their funds to a fixed menu of projects early in their existence [Grindley, 1988a]. In this sense, SEMATECH more closely resembled Japanese consortia such as VLSI, which relied on committees made up of member-company vice presidents and departmental managers to select research topics and allocate resources [Sakakibara, 1983], and to revise these decisions, over the life of the consortium.

Another characteristic of SEMATECH that distinguished it from many Western European consortia was the underlying strength, especially in product technology and design, of the target industry. Indeed, this underlying strength facilitated the commitment of resources by member firms to SEMATECH (in contrast, for example, to many SME firms, which lacked the necessary human or financial resources). Although serious, the competitiveness problems of the U.S. semiconductor industry were less severe than those of the European or U.K. IT industries at the inception of ESPRIT or Alvey [Grindley, 1988b]. SEMATECH could focus on a limited array of critical

²⁶ Japan's VLSI Project, in which joint research accounted for less than 20 percent of the total research budget, was the first Japanese research association to utilize joint research facilities. Even in the joint facilities, research was primarily conducted between pairs of firms rather than among all consortium members. Almost 90 percent of the Japanese consortia surveyed by Aldrich and Sasaki [1993] conducted the bulk of their research in member firms, well above the 44 percent of U.S. consortia reported to do so in this survey (p. 15).

²⁷ Although ESPRIT's design and structure incorporated the advice of a succession of European industry advisory groups, the impetus for the program came from senior EC policymakers, such as Etienne Davignon.

issues in process technology and equipment, but Alvey and ESPRIT, on the other hand, faced a need to virtually rebuild entire industries, which often had lost important complementary capabilities, such as marketing or firm-financed R&D, that were essential to the success of any cooperative R&D program. As we note below, SEMATECH has had serious difficulties in segments of the U.S. equipment industry that require much more than access to cooperative R&D.

SEMATECH's organizational structure also contrasts with that of its major privately funded counterpart in the U.S. electronics industry, MCC. SEMATECH was more successful at getting member firms to provide high-quality assignees than was MCC, a staffing policy that has facilitated the transfer of research results to SEMATECH member firms. The original mandates of MCC and SEMATECH also differ, inasmuch as SEMATECH long has focused on process development rather than basic research, and makes its research results available to all members.²⁸

EVALUATING SEMATECH

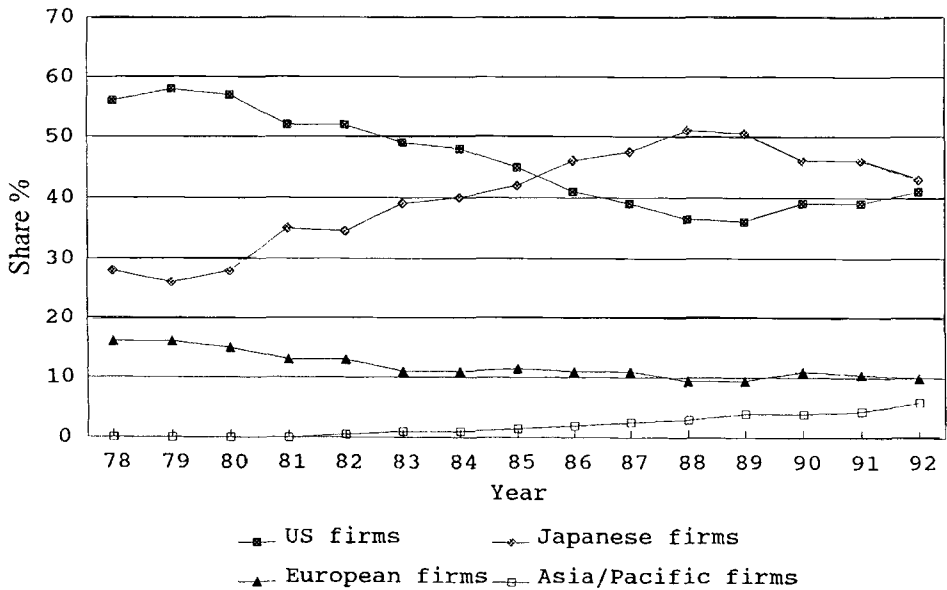
Evaluation Criteria

Is SEMATECH a success? Evaluating this consortium is difficult, for reasons that are common to many public-private technology partnerships—agreement on the goals of such programs and on evaluation criteria are lacking. Another problem is that the relevant time horizon for determining the effects of SEMATECH extends well beyond the consortium's five years of operations—the competitive effects of intrafirm R&D are realized only after a considerable lag, and it is reasonable to expect that consortia similarly take time to affect the economic fortunes of an industry. Even if the relevant quantitative data could be assembled, it is difficult to specify the counterfactual case against which one should compare the results of the public investment in SEMATECH. For example, what would have been the effects of \$500 million in federal grants to individual firms for the support of research on semiconductor manufacturing processes and equipment? Alternatively, should SEMATECH be compared against a situation in which no such investment in civilian semiconductor manufacturing research was made?

SEMATECH's adaptability yields another problem that is likely to arise in future consortia: Many of the consortium's original goals are no longer relevant to its evaluation. Flexibility in the SEMATECH research agenda is a strength that many European technology programs do not share. The consortium's ability to revise its goals and research agenda nevertheless creates difficulties of accountability and evaluation that cannot be ignored in a program that receives substantial public funds.²⁹ Although we cannot provide

²⁸ MCC projects often are selected and sponsored by groups of member firms, with research performed in-house or in member firms. In return for this effort, the project sponsors have the rights to any innovation for a year after its development; other MCC members then are granted access for an additional year, after which the rights are available for licensing outside MCC.

²⁹ This tension is aptly stated in a U.S. GAO report on SEMATECH: "While generally agreeing with our lessons learned, SEMATECH officials stated that our matters for congressional consideration neglect the necessity for flexibility in a consortium's response to changing environments,



Source: Dataquest

Figure 1. Worldwide semiconductor market share (by company base).

definitive answers to these questions, the widespread view of SEMATECH as a model for U.S. technology policy demands some consideration of its effects.

SEMATECH's Impact on the U.S. Semiconductor and SME Industries

SEMATECH has been widely credited with improving the competitiveness of U.S. semiconductor producers and equipment firms.³⁰ The turnaround in the fortunes of the U.S. semiconductor industry has coincided with the formation of SEMATECH. Since 1990 the U.S. share of the world market for semiconductor components has been increasing, and in 1992 the share returned to equality with that of Japanese firms (Figure 1).³¹ In 1992, a U.S. firm (Intel) regained the leading position in semiconductor sales (Table 2 shows the change in the leadership of the world market since 1980).³²

citing its decision in 1989 to alter its R&D program to give crucial attention to equipment suppliers. While we agree that a consortium needs flexibility, we continue to believe that a consortium needs to develop an initial operating plan with realistic objectives and milestones on the basis of a comprehensive assessment of the industry" [1992b, p. 13].

³⁰ For a representative comment, see "When Industrial Policy Works" [1992].

³¹ Preliminary Dataquest figures for 1993 estimate U.S. firms' share of world semiconductor component sales at 43.6 percent, slightly larger than the 40.2 percent share of Japanese firms [Siegman, 1993].

³² These market share data measure only reported sales of components and therefore exclude the "captive" production for internal use of firms such as IBM. Indeed, were these figures computed in terms of total output, thus including captive production of such U.S. firms as IBM and AT&T, U.S. firms would be shown to have maintained their dominance through the 1980s.

Table 2. Top 10 worldwide merchant semiconductor firms (1980–1992).

Rank	1980			1990			1992		
	Company	Sales (\$m)	Market share (%)	Company	Sales (\$m)	Market share (%)	Company	Sales (\$m)	Market share (%)
1	Texas Inst.	1,580	12.2	NEC	4,952	8.6	Intel	,5064	7.7
2	Motorola	1,110	8.5	Toshiba	4,905	8.5	NEC	4,976	7.6
3	Philips	935	7.2	Hitachi	3,927	6.8	Toshiba	4,765	7.3
4	NEC	787	6.1	Motorola	3,692	6.4	Motorola	4,635	7.1
5	National	747	5.7	Intel	3,135	5.5	Hitachi	3,902	5.9
6	Toshiba	629	4.8	Fujitsu	3,019	5.3	Texas Inst.	3,052	4.7
7	Hitachi	622	4.8	Texas Inst.	2,575	4.5	Fujitsu	2,583	3.9
8	Intel	575	4.4	Mitsubishi	2,476	4.3	Mitsubishi	2,307	3.5
9	Fairchild	566	4.4	Matsushita	1,945	3.4	Philips	2,108	3.2
10	Siemens	413	3.2	Philips	1,932	3.4	Matsushita	1,929	2.9
	Others	5,036	38.7	Others	24,943	43.4	Others	30,266	46.1
	Total*	13,000	100	Total	57,500	100	Total	65,587	100

Source: Dataquest.

* Totals include captive production.

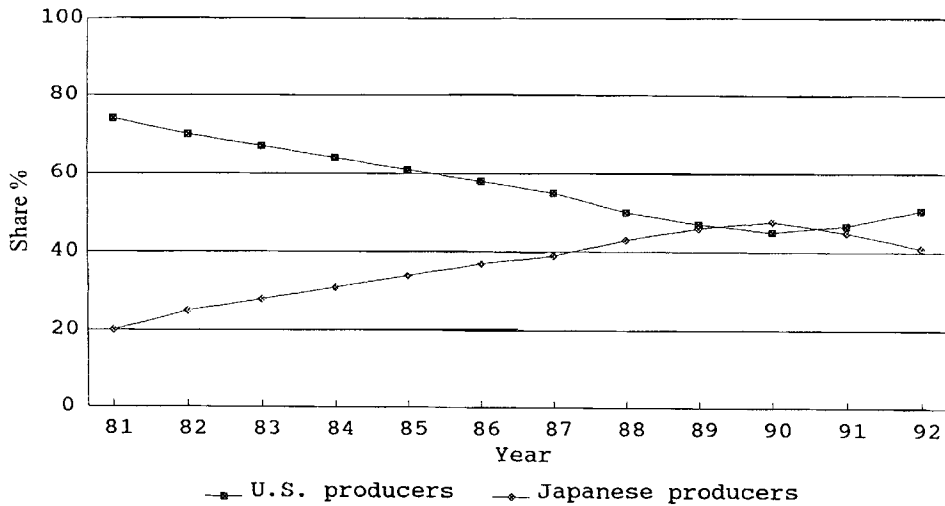


Figure 2. Worldwide SME marketshare (%) (by producer base).

This improvement in the performance of the U.S. semiconductor manufacturers may be based on gains in manufacturing performance that are attributable to SEMATECH-supported work on process technologies and SEMATECH support for improved equipment design, operation, and maintenance.³³ But there is little credible direct evidence that SEMATECH is responsible for such gains. The measured resurgence in U.S. producers' market share is based on the value of shipments and reflects the intense price competition during much of this period in the "commodity" segments of the industry, such as memory components, that has resulted from entry into memory production by large South Korean producers. The strong performance of U.S. firms in the more profitable microprocessor market has shielded their revenues somewhat from the effects of stiff price competition. Japanese and Korean firms thus far have failed to parlay the manufacturing skills acquired through production of high-volume memory chips into the design and manufacture of microprocessors that could threaten the position of U.S. firms.

U.S. semiconductor equipment suppliers have also improved their global market performance since 1991 (Figure 2), reversing a 20-year decline in their share of the world equipment market. In 1992, U.S. firms' share of the \$10.2 billion global market for semiconductor equipment rose to 50.5 percent from 46.7 percent in 1991, while Japanese firms' 1992 share fell to 41.0 percent from 44.9 percent.³⁴ The largest single supplier of semiconductor equipment

³³ According to VLSI Research, SEMATECH has improved yields of U.S. equipment to the point where the Japanese yield advantage has shrunk from 15 percent in 1985 to 9 percent in 1991 [U.S. GAO, 1992a].

³⁴ In 1992, U.S. SME producers picked up almost four points of market share in the U.S. and more than two points in Japan [Kelly, 1993, p. 27]. Most of this improvement, however, is due to the success of two firms: Applied Materials and Lam Research, whose worldwide sales increased 21 percent and 33 percent, respectively, over 1991 figures ("U.S. Equipment Vendors Gain," 1993).

in the 1992 world market was a U.S. firm, Applied Materials (see Table 3). A survey of SEMATECH member firms found that in 1991, they expected to purchase 40 percent of their new fab equipment from U.S. producers; in fact, however, members actually purchased 70 percent of their equipment from U.S. suppliers, which may also reflect SEMATECH-supported cooperation [SEMATECH, 1991].

Nevertheless, much of the reversal in the fortunes of U.S. SME firms during the past two years springs from the same source as the improvement in U.S. semiconductor manufacturers' performance—the decline in the economic fortunes of the Japanese producers who once dominated global markets for semiconductor memory chips.³⁵ The competition in memory chip production that has reduced the profitability of this product and reduced their share of global semiconductor revenues, combined with the broader Japanese business downturn, has led Japanese semiconductor firms to reduce their capital investments. Reductions in their investment plans have also reduced Japanese semiconductor firms' demand for new equipment, even as the growth of new entrants in Taiwan and Korea has enhanced Asian demand for non-Japanese equipment. A portion of the apparent improvement in U.S. SME firms' market share may reflect this shift in the regional distribution of new equipment demand and need not imply improved competitiveness in markets in which U.S. firms had lost market share to foreign firms.

An indication of this effect is given in Table 4, which compares changes during 1991–1992 in Japanese, U.S., and other regions' purchases of semiconductor manufacturing equipment with changes in SME sales by firms in each region, and with changes in the capital expenditures by each region's semiconductor firms.³⁶ The data indicate that changes in sales by U.S.- and Japan-based SME manufacturers broadly followed equipment purchase trends in their respective regional markets, trends that reflected contrasting growth rates in regional capital spending. U.S. equipment purchases in 1992 rose by 3 percent over their 1991 level, matching the 3 percent rise in U.S. SME firms' sales during this period. Japanese equipment purchases in 1992 fell by more than 25 percent from their 1991 level, while Japanese SME suppliers' revenues declined by 13 percent (U.S. semiconductor manufacturers' capital expenditures declined by 9 percent while Japanese capital expenditures declined by 29 percent during the 1991–1992 period). The increase in U.S. worldwide SME market share thus may be due largely to the shrinkage of the Japanese market for equipment, rather than a fundamental shift in world demand toward U.S. suppliers.³⁷ Given the regional shifts in demand for semiconductor manufacturing equipment, Japanese SME firms appear

³⁵ Fiscal year 1992 profits for Japan's four largest memory chip producers—NEC, Fujitsu, Toshiba, and Hitachi—declined 72 percent, 85 percent, 67 percent, and 44 percent, respectively, from their 1991 levels [Young, 1992; Doe, 1992]. Korean firms have garnered 20 percent of the 1-megabit DRAM market, and in 1992 Samsung replaced Hitachi as the world's largest producer of 4-megabit DRAMs [Young, 1992].

³⁶ Since the analysis combines data from different sources, and 1992 data are provisional, these results should be seen as indicative rather than definitive. For example, equipment purchase statistics include equipment only, while SME sales statistics include equipment and process materials.

³⁷ Historically, Japanese semiconductor manufacturers have purchased 80 percent of their equipment from Japanese suppliers [Thompson, 1993, p. 123].

Table 3. Top 10 worldwide SME suppliers (1979–1992).

Rank	1979			1989			1992		
	Company	\$m	HQ	Company	\$m	HQ	Company	\$m	HQ
1	Fairchild	111	USA	Tokyo Electron	634	Japan	Applied Matls.	794	USA
2	Perkin-Elmer	101	USA	Nikon	582	Japan	Tokyo Electron	677	Japan
3	Applied Matls.	54	USA	Applied Matls.	523	USA	Nikon	501	Japan
4	GCA	54	USA	Advantest	399	Japan	Canon	333	Japan
5	Teradyne	53	USA	Canon	384	Japan	Advantest	327	Japan
6	Varian	51	USA	General Signal	354	USA	Teradyne	278	USA
7	Tektronix	39	USA	Varian	335	USA	Varian	263	USA
8	Eaton	38	USA	Hitachi	210	Japan	Hitachi	258	Japan
9	Kulicke and Soffe	37	USA	Teradyne	200	USA	General Signal	231	USA
10	Balzars	34	Europe	ASM Intl.	187	Europe	{ASM Intl. {SVG	203 203	Europe USA
Market share									
USA		76%	Market share		USA		Market share		51%
Japan		16%	USA		Japan		Japan		41%

Note: Part of change in relative market shares due to yen appreciation over period.

Sources: *Tokyo Business Today* [1990], Davis [1990], and VLSI Research.

Table 4. Changes in semiconductor capital expenditure, equipment purchases, and SME market share (1991–1992).

	1991			1992			Growth 92/91* (%)
	\$bn	Share (\$)		\$bn	Share (%)		
Capital expenditure							
Japan	5.6	43		4.0	36		-29
USA/Canada	3.8	29		3.5	32		-8
Europe	1.2	9		1.1	10		-8
Asia/Pacific	2.4	18		2.4	22		0
Total	13.0	100		11.0	100		-15
Equipment purchases** (by market)							
Japan	4.1	46		3.0	37		-25
USA/Canada	2.7	30		2.8	34		3
Europe	0.9	11		1.0	13		11
Asia/Pacific	1.2	14		1.2	15		2
Total	8.9	100		8.1	100		-9
SME sales** (by HQ of supplier)							
Japan	4.8	45		4.2	41		-13
USA/Canada	5.0	47		5.2	51		3
Europe	0.6	5		0.6	5		-5
Asia/Pacific	0.3	3		0.3	3		-1
Total	10.7	100		10.2	100		-5

Sources: Capital Expenditure—Dataquest; Equipment purchases—Semiconductor Equipment and Materials International (SEMI) and Semiconductor Equipment Association of Japan (SEAJ); SME sales—VLSI Research.

* Growth 92/91 = [\$bn 1992—\$bn 1991]/[\$bn 1991] (using unrounded data).

** Equipment purchases include semiconductor equipment only. SME sales include equipment and process materials.

to have performed at least as well as U.S. firms. Once the Japanese economy recovers, therefore, so may the sales and market share of Japanese SME firms.

In one important segment of the U.S. SME industry, lithographic "steppers," SEMATECH programs have not prevented erosion in U.S. firms' competitiveness. This class of equipment accounts for a substantial share of SEMATECH spending on equipment development and improvement.³⁸ GCA, the inventor of the wafer lithographic stepper in 1970 and a major U.S. supplier, steadily lost market share during the past seven years, largely to foreign firms. GCA was an important participant in SEMATECH's equipment development programs, which supported roughly 20 percent of the costs of developing the firm's advanced XLS stepper [Dunn, 1993a].³⁹ SEMATECH also purchased 14 GCA ALS 200 I-line steppers and leased them to semiconductor manufacturers for beta-site testing and qualification, at a cost of \$19 million [Zipper, 1990; U.S. GAO, 1992a]. GCA technology was considered excellent, but the firm's business problems led to a March 1993 decision to suspend stepper production. Following lackluster bidding by domestic suitors, GCA was shut down in May 1993 [Dunn and Robertson, 1993]. (For a detailed examination of the issues surrounding GCA's demise, see Randazzese [1994].)

The only remaining large U.S. producer of lithographic equipment, SVG Lithography, announced in May 1993 that it had entered a joint venture with Canon of Japan for the development and production of the Micrascan stepper, another project that has received substantial SEMATECH support.⁴⁰ In a departure from its previous restrictions on foreign firms' access to its research projects, SEMATECH announced that it would continue to support SVGL's development of the Micrascan.⁴¹ In this important segment of the U.S. SME industry, SEMATECH's best efforts have not prevented the outright failure or financial decline of major U.S. producers. Indeed, the inability of SEMATECH to reverse the decline in this segment of the U.S. SME industry highlights the argument that technology alone is rarely sufficient to restore the competitiveness of firms or industries that lack critical, complementary personnel, marketing, managerial, and financial resources.

If the evidence on SEMATECH's direct contribution to improved economic competitiveness in semiconductor manufacturing or equipment is ambiguous, alternative measures of the consortium's performance suggest a more positive verdict. SEMATECH has met most of its revised objectives in the development of process technology, the supply of manufacturing equipment and collaboration between manufacturers, suppliers, and research centers. A 1992 survey of SEMATECH members [U.S. GAO, 1992a] found that most reported positive returns on their investments in SEMATECH, based on reductions in their costs of purchasing, operating, and maintaining U.S. equipment [Burrows, 1992]. The consortium also set out specific technical goals

³⁸ Between 1988 and 1992, SEMATECH devoted 38 percent of its \$287 million contract research budget to lithography [U.S. GAO, 1992a].

³⁹ SEMATECH reportedly devoted roughly \$70 million of joint development funds to GCA [Dunn, 1993b].

⁴⁰ According to Dunn [1993c], SVGL received \$30 million directly from SEMATECH and roughly \$20 million from IBM in support of Micrascan project development. IBM also contributed approximately \$80 million through equipment purchases.

⁴¹ William Spencer, the SEMATECH CEO, noted, "This is another step in the evolution of realizing that markets are no longer just the U.S., and if we're going to have strong companies, they have to sell worldwide" [Dunn, 1993c].

calling for scheduled demonstrations of the capability to produce smaller line-widths and larger wafers. These have also been accomplished essentially as planned. Phase 3, which demonstrated the production of 0.35 micron line-width integrated circuits on 8-inch wafers, was achieved in late 1992 [SEMATECH, 1992]. But the demonstration of this capability brought U.S. semiconductor manufacturing technology to parity with Japan, rather than enabling the U.S. semiconductor industry to surpass its Japanese competition, as members originally expected [U.S. GAO, 1992a].

The ambiguity of the link between SEMATECH as cause and improved competitiveness as effect will bedevil policymakers and evaluators of other public-private consortia. The experience of the U.S. lithographic stepper industry, however, suggests that collaborative support for technology development and improvement will not suffice in industries populated by relatively small, undercapitalized firms with severely constrained stocks of managerial and technical talent, especially when these firms face larger competitors with deeper pockets.

SEMATECH and the Role of Vertical Research Relationships Within Consortia

Although SEMATECH's emphasis on infrastructure has avoided many appropriability problems among its members, some of these problems have been shifted upstream to the SME industry. Just as SEMATECH's original agenda for research on manufacturing process technology raised concerns by chip manufacturers over sharing information with competitors, sharing between SME suppliers and SEMATECH member firms has raised concerns among suppliers that proprietary information will be disclosed to competitors. One SME supplier, Invax, has sued SEMATECH for allegedly revealing sensitive technical information to a competitor [Burrows, 1992].⁴² Although SEMATECH has been accused of favoritism in awarding equipment development and improvement contracts [Dunn, 1989], SME firms generally seem satisfied with SEMATECH's contract award process [Rice, 1989; Burrows, 1992]. The ability of SEMATECH to avoid major disputes with suppliers thus far may be due in large part to its exhaustive contract proposal review procedures, which include meetings by SEMATECH personnel with each bidding supplier to provide a detailed critique of their proposals' strengths and weaknesses.

SEMATECH's role in supporting equipment development and qualification also gives it considerable power to strengthen individual firms or technological alternatives in an industry faced with great uncertainty. This power, of course, means that important alternative paths of technological development may not be pursued—as we noted earlier, consortia can reduce diversity in technology development. Nevertheless, unlike earlier federally backed efforts to support the development of technologies with commercial applications [e.g., the SST], SEMATECH is to a large extent managed by the customers for these technologies. Its members have compelling incentives to maintain

⁴² In interviews, managers from some supplier firms expressed concern that if they fail to meet the equipment performance and schedule requirements associated with a SEMATECH project, the consortium might transfer sensitive information produced by the project to competitors in an effort to develop equipment that can meet performance goals. Another supplier feared that its proprietary technology would be leaked by SEMATECH member firms to a competitor through long-established procurement relationships between this competitor and specific member firms. There is no evidence, however, of any such leakage of sensitive information [interviews with SEMATECH, member firm, and supplier personnel, May–October 1993].

a diversified portfolio of technological alternatives and to revise their choices in response to new information. Thus far, SEMATECH also has diversified its risks in many technologies, pursuing contracts with more than one supplier and more than one engineering approach.

SEMATECH's important role within the U.S. equipment industry may affect this industry's structure. Since it has begun to assign considerable weight to firms' financial strength in awarding contracts, SEMATECH's equipment development program may benefit firms with greater resources. The consortium's centralized qualification procedure also means that SME firms working on equipment development or improvement contracts require SEMATECH approval to sell prototype tools. This policy arguably favors larger SME firms with the financial resources to weather the waiting period. Although these features of its programs should increase concentration in the U.S. SME industry, SEMATECH's effects on industry structure are uncertain. The number of mergers in the U.S. SME industry has grown since SEMATECH revised its research agenda,⁴³ but this trend reflects many factors other than the presence of SEMATECH. Indeed, one reason for the numerous recent mergers and acquisitions within the SME industry is the fact that a number of established U.S. firms have divested their SME divisions. High levels of restructuring thus coexist with a large number of small firms and continued fragmentation in industry structure. The pace of acquisition of U.S. SME suppliers by foreign firms also appears to have increased since SEMATECH changed its focus.⁴⁴

SEMATECH also faces a dilemma between its members' desire to capture the benefits of SME improvements and innovations and its objective of strengthening U.S. SME suppliers. As was noted above, SEMATECH has reduced its restrictions on sale to nonmembers of equipment developed with SEMATECH funding. SME suppliers opposed the restriction because it reduced their potential market, revenues, and profits. SEMATECH-financed equipment advances now diffuse throughout the global semiconductor industry more rapidly, increasing the benefits of SEMATECH research for SME suppliers and nonmember chip-makers. This shift in SEMATECH policy thus has increased the returns to "free-riding" by nonmembers in the U.S. and world semiconductor industries—SEMATECH's exclusion of foreign firms now imposes fewer obstacles on these firms' ability to benefit from SEMATECH research and results.

SEMATECH considers non-SEMI/SEMATECH member proposals and has on occasion awarded a contract to nonmembers, a policy that has increased tensions between the consortium and SEMI/SEMATECH.⁴⁵ As a result, some SME suppliers have questioned the value of membership in SEMI/SEMATECH [see Dunn, 1989]. Other equipment suppliers argue that SEMATECH occasionally acts as a barrier, rather than as a support, to com-

⁴³ Between 1987 and 1989, there were fewer than 60 acquisitions, mergers, or dissolutions in the U.S. SME industry. Between 1989 and 1991, there were more than 100 [Davis 1990; SEMI/SEMATECH 1991–1992].

⁴⁴ Foreign firms acquired 28 U.S. SME suppliers between late 1987 and the end of 1989 [Advisory Council, 1990]. Between January 1990 and June 1992, foreign firms acquired 53 U.S. suppliers [conversation with Miller Bonner, SEMATECH, July 1992].

⁴⁵ SEMATECH's willingness to entertain proposals from SME suppliers that are not members of SEMI/SEMATECH may be driven in part by political considerations. Some suppliers (not members of SEMI/SEMATECH) bristle at the idea of being required to pay a membership fee in order to have the right to bid for government-funded contracts [Dunn, 1989].

munication and information exchange between equipment suppliers and manufacturing firms. These equipment-firm managers wish to work directly with individual firms in equipment development projects, rather than working through SEMATECH, arguing that the costs of these development activities would be lower.

The efficiency gains from SEMATECH testing and qualification of equipment also have been questioned by some managers, who argue that individual purchasers of equipment still must conduct extensive debugging and beta-site activities within their production facilities. Some of this activity undoubtedly reflects differences among the manufacturing processes of member firms, as well as concerns by members over disclosure of sensitive aspects of their processes to other members. Other SME firms complain about SEMATECH's bureaucratic processes for setting standards and the length of time SEMATECH takes to develop and award contracts. But this risk must be balanced against SEMATECH's stabilization of equipment demand, which simplifies operations for SME suppliers.

Perhaps the greatest challenge for SEMATECH in structuring and managing this vertical research agenda results from the significant differences between its members and SME firms in size and in financial, as well as managerial, resources. Many of the SME firms are small, financially pressed, and staffed by few engineers and managers. Such firms may not have the resources to assign engineers to work at SEMATECH or at a member firm on a six-month equipment improvement project. Moreover, interviewees repeatedly stressed the point that the critical weaknesses of many equipment firms were financial and managerial, as much as technological.⁴⁶ Fundamental project management or marketing skills often were lacking. SEMATECH's efforts to provide training in total quality management (the PFTQ program) is one recognition of this problem, but more than training may be necessary to revive or sustain some equipment firms. In other instances, equipment suppliers were unable to absorb and apply sophisticated equipment design or experimental modeling techniques developed in SEMATECH projects in which they had participated. Many of SEMATECH's management and personnel policies, as well as its technology objectives, have been revised to address the very different nature of the technology development and transfer tasks when equipment firms, particularly small enterprises, are involved. But much remains to be done. These problems of collaboration between relatively large firms and small suppliers are likely to affect efforts at vertical collaboration in other U.S. industries, such as aerospace or automobiles.

Excluding Foreign Firms: Feasible or Desirable?

The domestic focus and restrictions on foreign membership of SEMATECH are difficult to sustain in a global industry like semiconductors. Tensions between national and international competitive realities are apparent among both equipment suppliers and semiconductor manufacturers, and

⁴⁶ SEMATECH managers have on occasion been confronted with serious financial crises in equipment firms and have been forced to consider financial bailouts for these enterprises. SEMATECH's mix of public and private financing, however, imposes serious restrictions on the consortium's use of funds for such purposes and reduces its flexibility.

SEMATECH members have become increasingly involved in joint ventures with large foreign rivals.⁴⁷

These foreign alliances have complex implications for U.S. suppliers. Although SEMATECH member firms control any flow of SEMATECH-developed technology to foreign partners within these alliances, international joint ventures weaken the effectiveness of the exclusion of foreign firms from SEMATECH. But these joint ventures also provide U.S. SME suppliers a means to gain access to foreign markets. If the American member of an alliance can persuade its foreign partner to use U.S. equipment in their venture, then the foreign chip-maker may continue to buy from the U.S. suppliers involved. The SME firms in SEMI/SEMATECH also have entered into a number of international joint ventures, such as that between Canon and SVGL, although the majority of these are for marketing or distribution rather than for research.⁴⁸ To the extent that these ventures among equipment firms encourage outward technology flows that reduce the lead time advantages that otherwise would accrue to SEMATECH member firms, their returns from participation in the consortium could decline. These international alliances obviously influenced SEMATECH's May 1993 exception to its policy of restricting even indirect participation by foreign firms.⁴⁹

The exclusion of foreign firms from a publicly supported technology development consortium also creates some difficulties for U.S. policy toward other governments' subsidization of high-technology industries. Although the U.S. government supports funding for technology development activities within the SEMATECH consortium, it opposes technology and product development subsidies extended by European governments to the Airbus consortium. Indeed, the policy governing government subsidies to technology development programs was a sensitive issue within the multilateral Uruguay Round of trade negotiations, and within the U.S. government [Stokes, 1993]. The apparent inconsistencies between this government's position on SEMATECH as a matter of technology policy and its position on subsidies as a matter of trade policy reflect broader problems of coordination between trade and technology policies that will figure prominently in industries other than semiconductors [Mowery, 1992b].

LESSONS OF SEMATECH

As we noted in the introduction, SEMATECH is likely to serve as a model for federal support of industrial technology in other sectors of the U.S. econ-

⁴⁷ For example, AMD-Fujitsu; AT&T-Mitsubishi; AT&T-NEC; IBM-Toshiba; Motorola-Toshiba; Texas Instruments-Hitachi.

⁴⁸ For example, Applied Materials-Komatsu; Eaton-Sumitomo; Lam Research-Tokyo Electronic; Varian-Tokyo Electronic.

⁴⁹ Any dramatic relaxation in this policy, however, such as opening SEMATECH to membership by foreign (particularly Japanese) semiconductor manufacturing firms, could jeopardize future federal funding. In response to SEMATECH's May 1993 announcement that it would continue to work with SVGL despite its alliance with Canon, the House Armed Services Committee criticized "the rationale of congressionally initiated investments if these investments do not result in jobs and industry in the U.S. In essence, in many respects the Department of Defense has become a research laboratory for aggressive foreign investors and cartels. It seems pointless to provide public funding primarily for the benefit of foreign firms" [Robertson and Dorsch, 1993].

omy.⁵⁰ The internationalization of such industries as aerospace or automobiles, in which large firms such as Boeing, General Electric, or Ford pursue extensive international collaborations with foreign counterparts and suppliers, has intensified competitive pressure on U.S. supplier firms in these and other industries. Many of these supplier firms, like U.S. SME firms, are relatively small and have serious technological, managerial, and financial weaknesses. In aerospace, many of these firms also are important sources of defense, as well as civilian, components. Can the vertical-collaboration model of SEMATECH strengthen U.S. suppliers and sectoral competitiveness in these and other U.S. industries? This section considers some lessons of SEMATECH's structure, agenda, and management.

Short-Term Focus

Like other industry consortia in Japan and the U.S., SEMATECH now focuses on technology development, rather than fundamental research. The shift in SEMATECH's research agenda from leading-edge, proprietary technology to near-term support for the industry's supplier infrastructure resolved conflicts over the research agenda, at the cost of losing several smaller members. SEMATECH's experience supports the argument that consortia are likely to be most effective in supporting the improvement and adoption of technology, rather than focusing on long-term research.

Its near-term focus, however, means that SEMATECH-based R&D cannot substitute for the longer-term R&D activities that government, universities, and industry have long performed. SEMATECH is dealing with a number of critical aspects of the technological exploitation of fundamental research, and especially with the adoption of technological advances. But it cannot substitute for broader support for the scientific and technological infrastructure of the U.S. economy. Nor, in the nature of the case, is much of this support likely to flow from private sources.

SEMATECH's efforts to improve the semiconductor industry's equipment supply infrastructure are insufficient by themselves to restore the competitiveness of this industry, but they have aided equipment and manufacturing firms that have strong financial and management resources. But this "generic" focus appears to be most beneficial to firms with strong in-house technological capabilities, a recurrent theme in research on consortia [Mowery, 1983]. This issue is especially important in SEMATECH's efforts to revive or strengthen U.S. SME firms, which often lack the in-house resources to exploit the results of SEMATECH projects. For example, strong firms are better able to provide the assignees that are essential to SEMATECH's operations. Where an industry's problems reflect broader weaknesses, which may be the case in the European IT industry (and, possibly, the U.S. lithographic

⁵⁰ U.S. public/private consortia established since SEMATECH includes the Multichip Module (initial budget, funded from public and private sources, \$60 million for an unspecified time period), Flat Panel Display (\$20 million from public and private sources for first year), Amtex (\$25 million from the Department of Energy and \$25 million from the U.S. textiles industry over five years), and the Advanced Battery (\$260 million from public and private sources over four years) Consortia. More than a dozen dual-use technology consortia are included as part of a new \$550 million Defense Technology Conversion proposal. Proposals include consortia to support the development of huge memory chips and parallel computers [Robertson, 1993].

stepper industry), incremental programs such as SEMATECH may be insufficient. Indeed, SEMATECH may accelerate rationalization or even a shakeout within the U.S. semiconductor equipment industry.

The free-rider problems that might otherwise undercut its feasibility are reduced somewhat for SEMATECH by the fact that its members account for the bulk of the U.S. semiconductor industry's output. Individual members derive sufficient benefits from their involvement in equipment development, standard-setting, and technology forecasting to maintain their commitment to the consortium. Although several of the original members withdrew from the consortium, SEMATECH has passed a critical "market test" of viability, as 11 of the original member firms thus far have maintained their contributions.

Freestanding Organization

Formulating and implementing a technology strategy, and especially revising this strategy, are more feasible in a centralized organization than in an umbrella consortium of independent projects. As we noted earlier, SEMATECH's relatively centralized structure appears to have enhanced the flexibility and responsiveness of its research agenda, especially by comparison with Western European consortia. But a centralized structure may not be well suited to the needs of proprietary technology development. Collaboration for these purposes may be best achieved in projects with fewer members. This consideration, along with the force of distributive political motives, may account for the umbrella structure of European programs.

Informed Management and Industry Support

SEMATECH's ability to shift its research agenda depended crucially on industry involvement in program management. Such involvement was ensured by industry's role in initiating the SEMATECH program and in financing the consortium. The importance of industry cofunding and participation has long been emphasized in evaluations of U.S. and other nations' technology programs, and the SEMATECH experience supports this conclusion.⁵¹

Although they did enlist industry funds, and ESPRIT reflected pressure from private-sector groups, programs such as Alvey and ESPRIT were motivated more by broad economic and political concerns than by industry initiative; they had equivocal support at best from industry. These programs frequently were unable to recruit leading industry executives to administer them. Others abandoned centralized management for any function other than administration. Despite efforts to make these the "industry's policy" rather than "industrial policy," industrial commitment was often half-hearted. Weak industry involvement in strategic planning meant that projects were selected according to obscure criteria and priorities were difficult to change. As we have noted, maintaining broad involvement by its member firms in project selection and oversight is an important continuing objective of SEMATECH.

⁵¹ For an early evaluation of federal demonstration projects that stresses the importance of financial commitments from industry, see Baer et al. [1977].

Is or Was Government Sponsorship Necessary?

Government support for some of the costs of establishing SEMATECH helped convince major U.S. semiconductor manufacturers to collaborate. As the major financier, the federal government made the program credible. Public financing also provided some guarantee of continuity for the first five years of operation and, at least implicitly, indicated government cooperation on other policy issues. By improving cooperation among manufacturing firms and (even more important) between semiconductor manufacturers and their U.S. suppliers of equipment and materials, SEMATECH had an important catalytic effect on this U.S. industry. SEMATECH's role within the U.S. semiconductor industry was itself catalyzed by the federal financial contribution.

Although federal funding has been important, must such funds continue to cover one half of SEMATECH's budget, and should these funds flow from the U.S. defense budget? In order to retain industry involvement in and commitment to SEMATECH, its member firms must retain primary control over the consortium's research agenda. This industry control of the SEMATECH research agenda favors projects with relatively near-term benefits, that in some cases yield little "spin-on" to military applications. Now that the consortium is established, future federal funding might be earmarked for specific fields of research (e.g., environmental protection), or awarded on the basis of competitive bids for research contracts.⁵² But efforts by federal sponsors to tighten control of SEMATECH's research agenda for agency missions may have a chilling effect on industry contributions of funding and personnel.

SEMATECH now performs some of the functions within the U.S. semiconductor manufacturing industry that in other industries are carried out by trade associations or standard-setting bodies. The economic case for federal financial support for SEMATECH and other consortia is strongest in research areas characterized by high uncertainty, lengthy time horizons, and substantial (technological or economic) spillovers and external effects that are rebound primarily to the benefit of U.S. firms and taxpayers. To the extent that SEMATECH's research agenda is dominated by routine equipment qualification, standard-setting, and support for suppliers' management skills, the case for large federal subsidies will be weakened.

A Model for Other U.S. Industries?

SEMATECH has inspired numerous proposals for collaborative R&D in other U.S. industries. The feasibility and ultimate success of such collaboration in these other sectors will be influenced by three sets of factors: (1) the structure of the consortia; (2) the political or economic expectations imposed on the consortia by sponsors; and (3) the "fit" between the activities of consortia and the competitive problems within an industry.

⁵² In 1992, DARPA proposed a shift away from institutional support for SEMATECH, instead allocating \$80 million annually to specific microelectronics manufacturing research projects; SEMATECH and other entities would compete for such research contracts [U.S. GAO, 1992b, p. 11]. Such a change would enhance federal agencies' control over SEMATECH's research agenda, at the cost of reducing the flexible research agenda and industry involvement that currently characterize SEMATECH. Although congressional appropriators earmarked \$20 million in annual federal funds for environmental research in the fiscal 1993 SEMATECH budget, they have not supported any additional changes in DARPA funding policies toward SEMATECH.

SEMATECH provides some reasonably clear guidelines for structuring and managing consortia in the U.S. context, especially when compared with consortia in other industrial economies. These structural characteristics have been extensively discussed in this section, and include requirements for sponsorship, management, staffing, and substantial funding from industry; substantial flexibility and autonomy; and the definition of a feasible agenda. SEMATECH's experience suggests that consortia are well-suited to supporting collaboration between users and suppliers that focuses on relatively near-term technology development, support for technology adoption, and improved communication and cooperation between users and suppliers.

But this agenda is likely to be one focusing on incremental improvement and diffusion, rather than "frontier" research. Public and private sponsors of industry-led consortia must recognize that these entities are not well-suited to long-range research and adjust their expectations accordingly. Public and private sponsors also must recognize the critical economic importance of foreign partners, markets, and suppliers to the U.S. firms that are likely to participate in such consortia. These economic realities mean that exclusion of foreign participants from domestic U.S. R&D consortia is likely to be infeasible, counterproductive, or both.

As we have noted above, SEMATECH has been more successful in strengthening U.S. SME and semiconductor firms that were already relatively strong. In particular, SEMATECH has encountered serious difficulties in overcoming the effects of the serious financial or managerial weaknesses that are widespread in the supplier tier of the U.S. semiconductor and other high-technology industries. In addition, the demands of collaboration between relatively small suppliers and well-established users of equipment differ substantially from those associated with collaboration among large, research-intensive firms. In other words, vertical collaboration in other industries is likely to prove more effective for supplier firms with significant in-house management, financial, and technological strength. Indeed, consortia in other industries may have to develop innovative techniques of financial or managerial assistance to address the broader competitive weaknesses of the supplier firms in many of these sectors.

CONCLUSION

SEMATECH's original objectives sought to elevate U.S. semiconductor manufacturing to a world leadership position, without specifying the means to achieve this objective. The consortium's ability to recognize the need for change in its research agenda and act on it was aided by the program's structure. This flexibility was crucial to SEMATECH, but its successful exploitation rested on the involvement of experienced industry managers in the consortium's operations. The involvement of industry executives in SEMATECH management meant that problems with the consortium's research agenda were recognized and effectively communicated to its members. SEMATECH's autonomy facilitated its response to these problems. SEMATECH managers were sufficiently acquainted with the industry to develop an alternative agenda to partially reach its initial goals. Continued industry involvement is essential to maintaining SEMATECH's effectiveness in developing and transferring technology to member and SME firms.

The coordination of projects in most European umbrella consortia differs considerably and more often than not has been purely administrative. Political and budgetary constraints also require early and detailed specifications of projects in these consortia, reducing flexibility in the research agenda. Projects often have been selected and monitored by administrators and committees remote from the industry, and change in program goals or structure has been especially difficult because of the political considerations that influence the establishment, structure, and research agenda of these programs.

Nevertheless, the very flexibility and autonomy that have aided SEMATECH's operations create dilemmas for public officials charged with (partial) funding and oversight of consortia. The broad goals of such undertakings, such as the "achievement of world leadership," generally are affected by so many influences other than collaborative R&D, and require so much time for their achievement, that they are unrealistic yardsticks for evaluation. In addition, a concept like "industry competitiveness" is very difficult to measure objectively. An alternative criterion for evaluation focuses on the specific tactics adopted to achieve broad goals. In the case of SEMATECH, one might include its technical goals, as well as its focus on user-supplier cooperation, as tactics. If these are agreed on at the inception of such a program, evaluation can focus on monitoring them. But if these tactics themselves are subject to significant shifts, evaluation becomes much more difficult. Evaluation problems are intensified still further when the goals of public and private funders of a consortium (for SEMATECH, respectively ARPA and SEMATECH's members) diverge somewhat.

Does the design of SEMATECH ensure its cost-effective operation? Its funding depends on continued support from industry, which provides incentives for efficiency in operation. But the efficiency of the processes that underpin the allocation of federal funds to SEMATECH could be improved significantly. SEMATECH's federal funding is not based on any formal competition with other proposals for industry-led consortia (in contrast, for example, to the Commerce Department's Advanced Technology Program) or peer evaluation of its research performance. A requirement, for example, that SEMATECH and other federally financed consortia rely exclusively on competitive contract awards for specific research projects might well shift their research to serving the needs of federal agencies, rather than those of industry. But some process of competition among proposals for initial and continuing "institutional" support from the federal government for industry-led consortia is desirable.

The internal debates over the SEMATECH research agenda suggest that horizontal collaboration among firms on process technologies is difficult in industries in which product innovation depends on process innovation. SEMATECH's vertical research agenda may in fact produce economic benefits for a larger group of member and nonmember firms, as research results now are released and diffuse more rapidly among both semiconductor producers and SME firms. This tendency gives rise to a dilemma that may affect SEMATECH's long-term viability. The current costs of SEMATECH membership are substantial; if additional member firms leave the consortium, a smaller number of semiconductor manufacturers will have to underwrite the costs of research benefiting the entire industry (including, in some cases, the non-U.S. semiconductor industry). Should this possibility be realized, the consortium might well be pressured by its members to develop a

more proprietary policy toward disclosure of research results. But any more restrictive policy would undercut the benefits of SEMATECH to the U.S. equipment industry. An alternative to increases in membership costs and shrinkage in the member-firm population is expansion in SEMATECH's domestic membership and a reduction in membership fees.⁵³ A broader SEMATECH membership base would reduce the conflicts between the financial burdens of membership and disclosure of research results. But SEMATECH's centralized structure could prove unwieldy in the face of the substantial growth in membership that might result from this change in policy.

Yet another resolution of this dilemma would open SEMATECH to foreign firms, expanding the number of member firms willing to underwrite the current membership fees and able to benefit from the current distribution of SEMATECH research results and benefits. But this policy shift would require extensive negotiation with the federal agencies currently supporting a large share of SEMATECH's operating expenses. In the long run, however, it is difficult to see how one of these scenarios can be avoided if SEMATECH is to remain robust.

Perhaps because SEMATECH's research agenda originated from and was subsequently modified by industry personnel, many of its aims have been achieved. SEMATECH sought to strengthen a relatively narrow array of capabilities in which the U.S. semiconductor industry was weak in the late 1980s, rather than attempting to reconstruct an entire industry. This approach is reminiscent of Japanese consortia, which aim at specific areas of industry weakness but leave the rest to the firms. This narrower focus also contrasts with European consortia, many of which have unrealistically attempted to solve an abundance of problems in weak industries. But SEMATECH has not been equally beneficial to all of its original or current members, nor have its activities strengthened all segments of the U.S. semiconductor equipment industry. Programs like SEMATECH can complement the technical and management strengths of member firms, but they cannot substitute for the complete absence of these strengths. This characteristic may limit the effectiveness and feasibility of industry-led consortia in industries with little or no industry-funded R&D.

Another element of similarity between SEMATECH and many Japanese

⁵³ When SEMATECH was established, the consortium adopted rules under which a new member would have to pay all past dues that it would have paid had it joined at SEMATECH's inception. This policy was designed to discourage potential members from adopting a wait-and-see attitude during SEMATECH's early years. SEMATECH's Board of Directors has agreed in principle to eliminate its "back-dues" requirement after 1992 [U.S. GAO, 1992a]. Nevertheless, few nonmember firms have the resources to make membership in SEMATECH worthwhile as long as the \$1 million minimum annual dues requirement remains.

One way to increase membership, as suggested by the U.S. GAO [1992b, p. 9], would establish a lower-cost "associate" membership through which smaller firms could participate in a subset of SEMATECH activities. This is similar to the structure adopted by MCC in 1984 in an attempt to expand its membership. As of July 1993, MCC had 47 associate members in addition to its 22 shareholder members (in March 1993, MCC established "small business associate" and "university affiliate" memberships as well; as of July 1993, the consortium had 15 of the former and 11 of the latter). A reduced fee option might attract new members to SEMATECH—Cypress Semiconductor CEO T. J. Rodgers acknowledges that "if [SEMATECH] cut their dues in half, we would take a very hard look at joining" [Burrows, 1992]—but SEMATECH has not yet altered its dues structure.

research consortia is this consortium's focus on nearer-term projects of a generic character; sensitive, proprietary technologies are developed by firms privately and in limited alliances. By contrast, the European programs that have moved toward shorter-term projects have become more proprietary. SEMATECH, MCC, and Western European research consortia all have responded to industry demands for short-term results that are closer to the market and easier to transfer to the firms. All of these cooperative research efforts have shifted away from long-term research, which may not be appropriate for consortia and better left to individual firms, university–industry cooperative programs, or public research institutes. SEMATECH seems likely to continue to move in this direction, and increasingly stresses near-term deliverables in its aim to improve the returns on members' financial contributions. From a national perspective, industry-led consortia are insufficient by themselves to maintain the infrastructure of high-technology industries. Their research efforts must be supplemented by longer-range research and training activities that by their very nature can be funded only from public sources and conducted in relatively "open" institutional environments that emphasize disclosure and dissemination [see Dasgupta and Davis, *in press*].⁵⁴

The U.S. semiconductor industry is strong technologically and at the frontier in areas that depend on rapid product innovation. SEMATECH has sought to improve an area in which U.S. semiconductor manufacturing firms lagged behind Japanese competitors, namely, the weakness of vertical user–supplier relationships. This research agenda has more to do with "catching up" with global best practice than with leading the next generation. SEMATECH may be a model for other catch-up situations, but its history suggests that such a program can work only in an industry that retains strength in other technical and nontechnical areas.

APPENDIX A: CONSORTIA ABBREVIATIONS

Japan

5G	Fifth Generation Computer Project
FED	R&D Association for Future Electronic Devices
ICOT	Institute for New Generation Computer Technology
MITI	Ministry of International Trade and Industry
NFE	New Function Elements Project
OEIC	Opto-electronics Integrated Circuits Project
VLSI	Very Large Scale Integration

United Kingdom

DTI	Department of Trade and Industry
MOD	Ministry of Defence
SERC	Science and Engineering Research Council

Europe

AIM	Advanced Informatics for Medicine
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⁵⁴ SEMATECH retains links to academic research institutes through its university-based "centers of excellence" and through the Semiconductor Research Corporation, so as to maintain contact with frontier research.

BCR	Meteorological and Chemical Measurement Research
BRIDGE	Biotechnology Research
BRITE	Basic Research in Industrial Technologies
ELDO	European Launcher Development Organization
ESA	European Space Agency
ESPRIT	European Strategic Program for Research and Development in Information Technologies
ESRO	European Space Research Organization
EURAM	European Research in Advanced Materials
EUREKA	European Research Technology Program
FAR	Fisheries Advanced Research
FLAIR	Food Technologies Research
JESSI	Joint European Submicron Silicon Project
STAR	Special Telecommunications Action for Regional Develop- ment
RACE	R&D in Advanced Communications for Europe
United States	
DARPA	Defense Advanced Research Projects Agency
DOD	Department of Defense
MCC	Microelectronics and Computer Technology Corporation
NCMS	National Center for Manufacturing Sciences
SDI	Strategic Defense Initiative
SEMATECH	Semiconductor Manufacturing Technology Corporation
SEMI	Semiconductor Equipment and Materials International
SIA	Semiconductor Industry Association
SRC	Semiconductor Research Corporation
VHSIC	Very High Speed Integrated Circuit Project

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